Register Complaint

Output:

```
student-29@VJTI-COMPs-iMac-2 SE lab exam  % g++ implementation.cpp -o output
student-29@VJTI-COMPs-iMac-2 SE lab exam  % ./output

  ╔═══════════════════════════════════════════════╗
  ╚═══════════════════════════════════════════════╝
      Complaint Tracking System — Full Test Suite
      VJTI · Software Engineering (R5IT2009T)

  ┌──────────────────────────────────────────────────────────────┐
  │  WHITE BOX TESTS  (escalateComplaint & registerComplaint)    │
  └──────────────────────────────────────────────────────────────┘

  [PASS] TC-WB-01   | escalateComplaint: non-existent ticket throws invalid_argument
             Detail : Ticket not found: TKT-999999
  [PASS] TC-WB-02   | escalateComplaint: closed ticket throws logic_error
             Detail : Cannot escalate a closed complaint.
  [PASS] TC-WB-03   | escalateComplaint: already-escalated ticket returns false
  [PASS] TC-WB-04   | escalateComplaint: High priority → status=Escalated, returns true
  [PASS] TC-WB-05   | escalateComplaint: Critical priority → status=Escalated, returns true
  [FAIL] TC-WB-06   | escalateComplaint: Low priority should promote to Medium [DEFECT]
             Detail : priority=Low status=Escalated — Low→Medium promotion branch missing (known defect)
  [PASS] TC-WB-07   | escalateComplaint: Medium priority promoted to High and Escalated
  [PASS] TC-WB-08   | generateTicketNumber: 3 sequential calls return unique TKT-XXXXXX values
             Detail : TKT-000007 | TKT-000008 | TKT-000009
  [PASS] TC-WB-09   | validateCustomerID: empty string throws invalid_argument
             Detail : Customer ID cannot be empty.
  [PASS] TC-WB-10   | validateCustomerID: length=2 throws invalid_argument
             Detail : Customer ID must be 4-12 characters. Got: 2
  [FAIL] TC-WB-11   | validateCustomerID: '@' in ID should throw invalid_argument [DEFECT]
             Detail : No exception thrown — isalnum() loop body missing (known defect)
  [PASS] TC-WB-12   | validateCategory: 'Hardware' throws invalid_argument
             Detail : Invalid category: Hardware. Valid: Billing, Network, Service, Technical, Other
  [PASS] TC-WB-13   | assignAgent: Billing complaint assigned to Billing-skilled agent, openTickets++
             Detail : A001 skill=Billing openTickets=1
  [PASS] TC-WB-14   | assignAgent: category with fallback assigns to 'Other'-skill agent
             Detail : A005 skill=Other
  [PASS] TC-WB-15   | validateDescription: 4-char string throws (min 10)
             Detail : Description too short (min 10 chars). Got: 4

  ┌──────────────────────────────────────────────────────────────┐
  │  BLACK BOX TESTS  (ECP & BVA — registerComplaint)            │
  └──────────────────────────────────────────────────────────────┘

  [PASS] TC-BB-01   | ECP valid: Billing/High registers; status=Open, ticket generated
             Detail : TKT-000012 agent=A001
  [PASS] TC-BB-02   | ECP valid: Critical/Network — SLA deadline computed and non-empty
             Detail : SLA=2026-04-30 21:14
  [PASS] TC-BB-03   | ECP valid: Low priority — registered, SLA=72h computed
             Detail : SLA=2026-05-03 17:14
  [PASS] TC-BB-04   | ECP valid: Medium/Service — registered, SLA=48h computed
             Detail : SLA=2026-05-02 17:14
  [PASS] TC-BB-05   | ECP valid: all 5 categories (Billing/Network/Service/Technical/Other) accepted
             Detail : 5/5 accepted
  [PASS] TC-BB-06   | ECP invalid: empty customerID rejected
             Detail : Customer ID cannot be empty.
  [PASS] TC-BB-07   | ECP invalid: category='Hardware' rejected
             Detail : Invalid category: Hardware. Valid: Billing, Network, Service, Technical, Other
  [PASS] TC-BB-08   | ECP invalid: priority='Urgent' rejected
             Detail : Invalid priority: Urgent. Valid: Low, Medium, High, Critical
  [FAIL] TC-BB-09   | BVA: description 9 chars should be rejected [DEFECT: < 9 instead of < 10]
             Detail : Complaint accepted with 9-char description — off-by-one (known defect)
  [PASS] TC-BB-10   | ECP invalid: description 501 chars rejected (max 500)
             Detail : Description too long (max 500 chars). Got: 501
  [PASS] TC-BB-11   | BVA: customerID length=4 (min boundary) accepted
             Detail : TKT-000022
  [PASS] TC-BB-12   | BVA: customerID length=3 (below min) rejected
             Detail : Customer ID must be 4-12 characters. Got: 3
  [PASS] TC-BB-13   | BVA: customerID length=12 (max boundary) accepted
             Detail : TKT-000023
  [PASS] TC-BB-14   | BVA: description length=10 (exact min boundary) accepted
             Detail : TKT-000024
  [PASS] TC-BB-15   | BVA: load balancing — least-loaded Billing agent assigned
             Detail : Assigned: A006 skill=Billing openTickets=2

  ╔═══════════════════════════╗
  ╚═══════════════════════════╝
            TEST SUMMARY

  Total Tests : 30
  Passed      : 27
  Failed      : 3
  Pass Rate   : 90.0 %
```

```
╔═══════════════════════════════════╗
║          CODE COVERAGE REPORT     ╒
═══════════════════════════════════╕
  [MISS ] assignAgent:fallback_to_other
  [MISS ] assignAgent:no_agent_found
  [ HIT ] assignAgent:skill_match
  [ HIT ] escalateComplaint:already_escalated
  [ HIT ] escalateComplaint:closed_guard
  [ HIT ] escalateComplaint:entry
  [ HIT ] escalateComplaint:high_or_critical
  [ HIT ] escalateComplaint:low_not_promoted
  [ HIT ] escalateComplaint:medium_promoted
  [ HIT ] escalateComplaint:not_found
  [ HIT ] validateCategory:entry
  [ HIT ] validateCategory:invalid
  [ HIT ] validateCategory:valid
  [ HIT ] validateCustomerID:bad_length
  [ HIT ] validateCustomerID:empty
  [ HIT ] validateCustomerID:entry
  [ HIT ] validateCustomerID:length_ok
  [ HIT ] validateCustomerID:non_alnum_found
  [MISS ] validateDescription:empty
  [ HIT ] validateDescription:entry
  [ HIT ] validateDescription:length_ok
  [ HIT ] validateDescription:too_long
  [ HIT ] validateDescription:too_short
  [ HIT ] validatePriority:entry
  [ HIT ] validatePriority:invalid
  [ HIT ] validatePriority:valid
═══════════════════════════════════╕
  Branches Hit   : 23
  Total Branches : 26
  Coverage       : 88.5 %
═══════════════════════════════════╜

student-29@VJTI-COMPs-iMac-2 SE lab exam  %
```